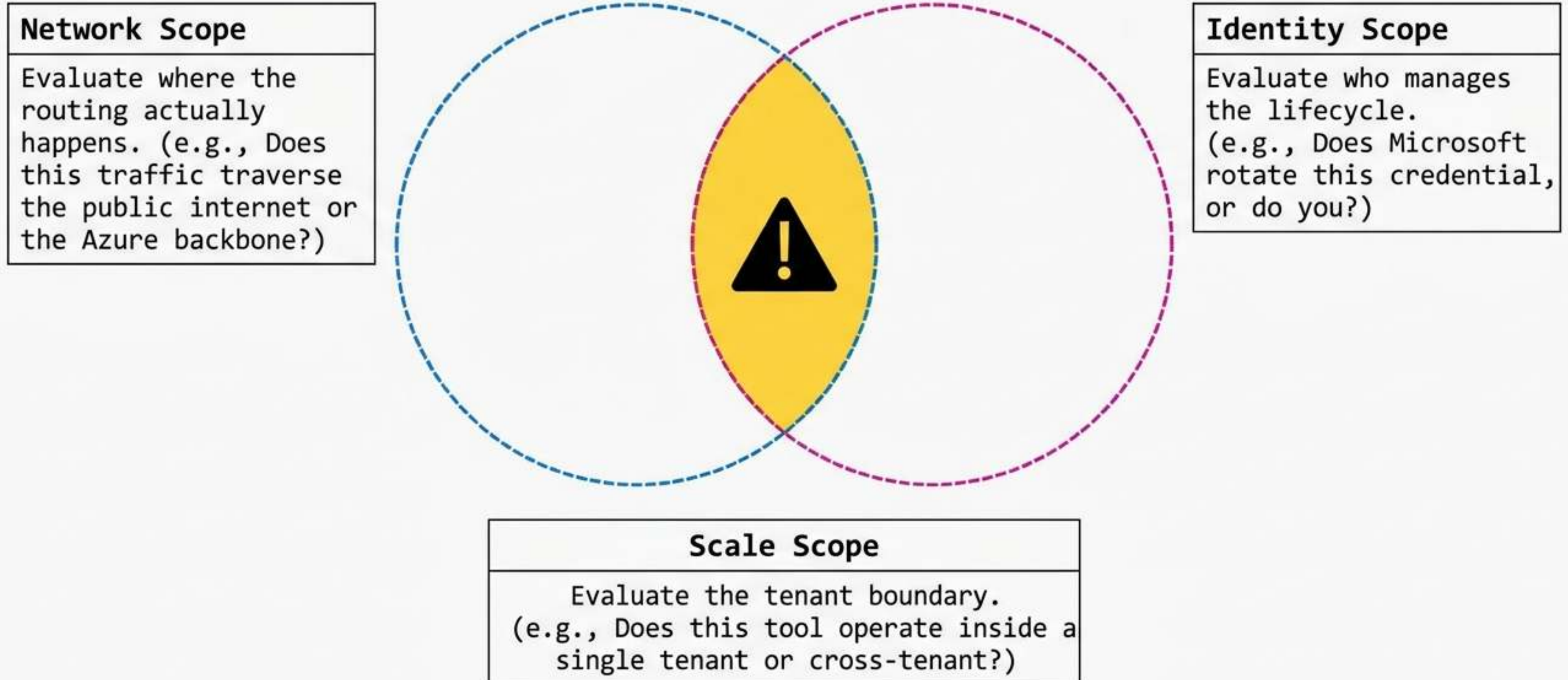


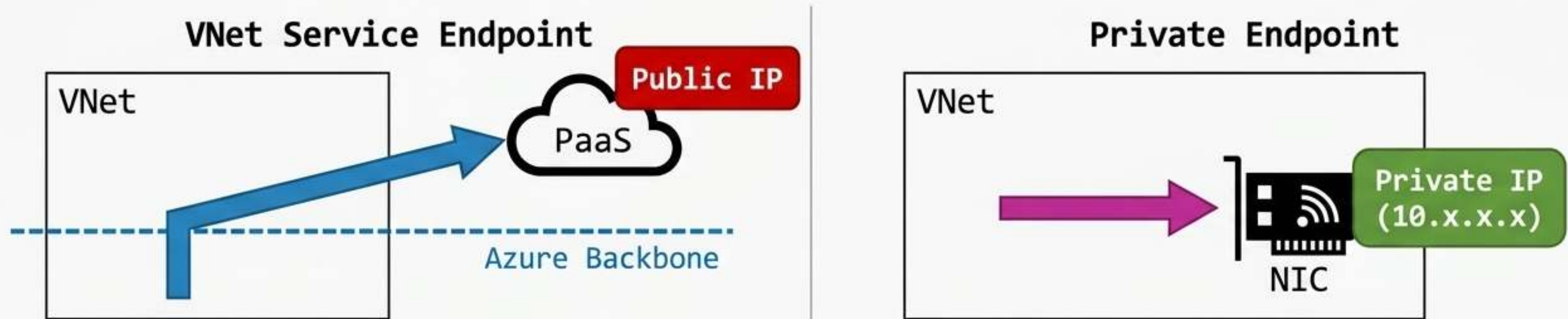
AZ-500 Deciphered: The Ultimate Comparison Matrix

Master the architectural pivot points between Azure's most confusing security services

The Exam Tests the Boundaries Between Services

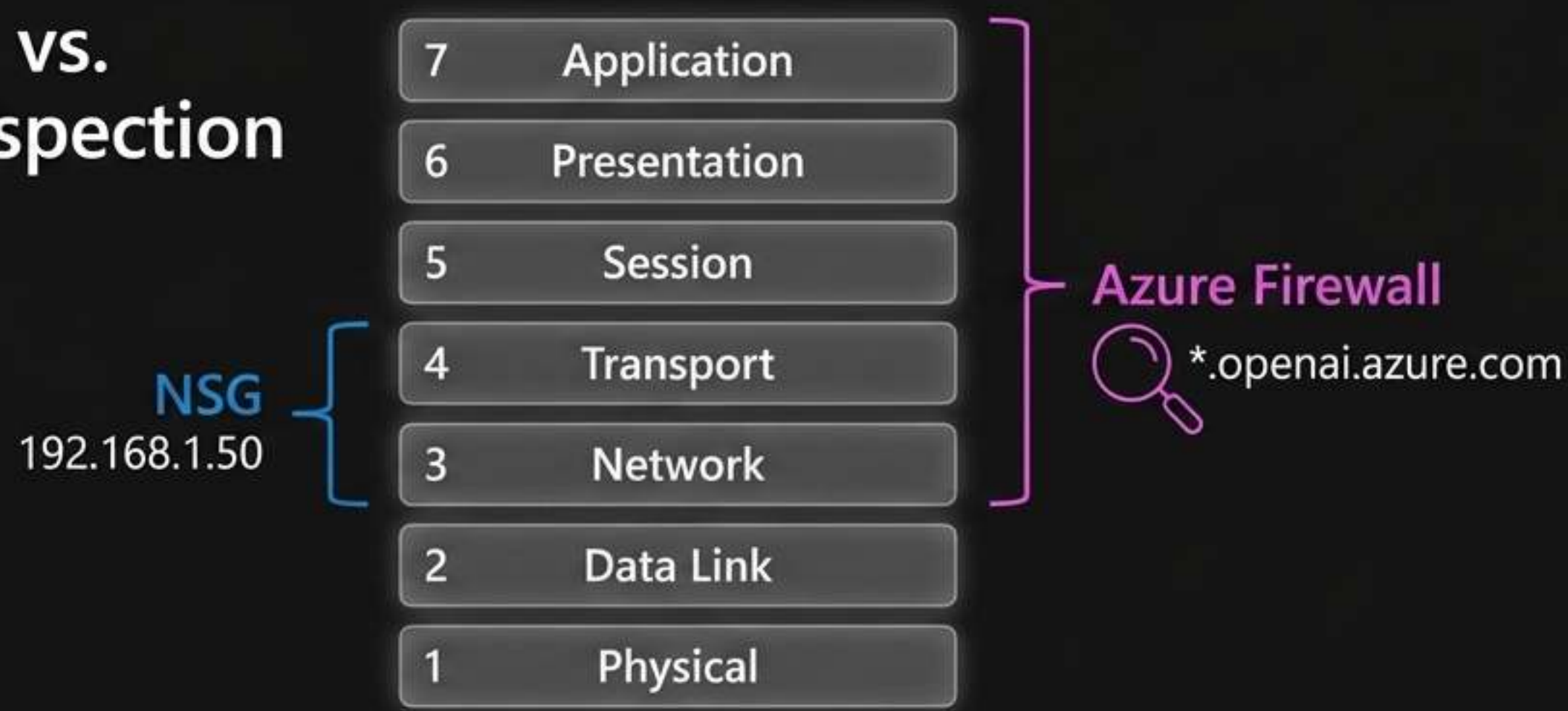


The Battle to Eliminate the Public IP



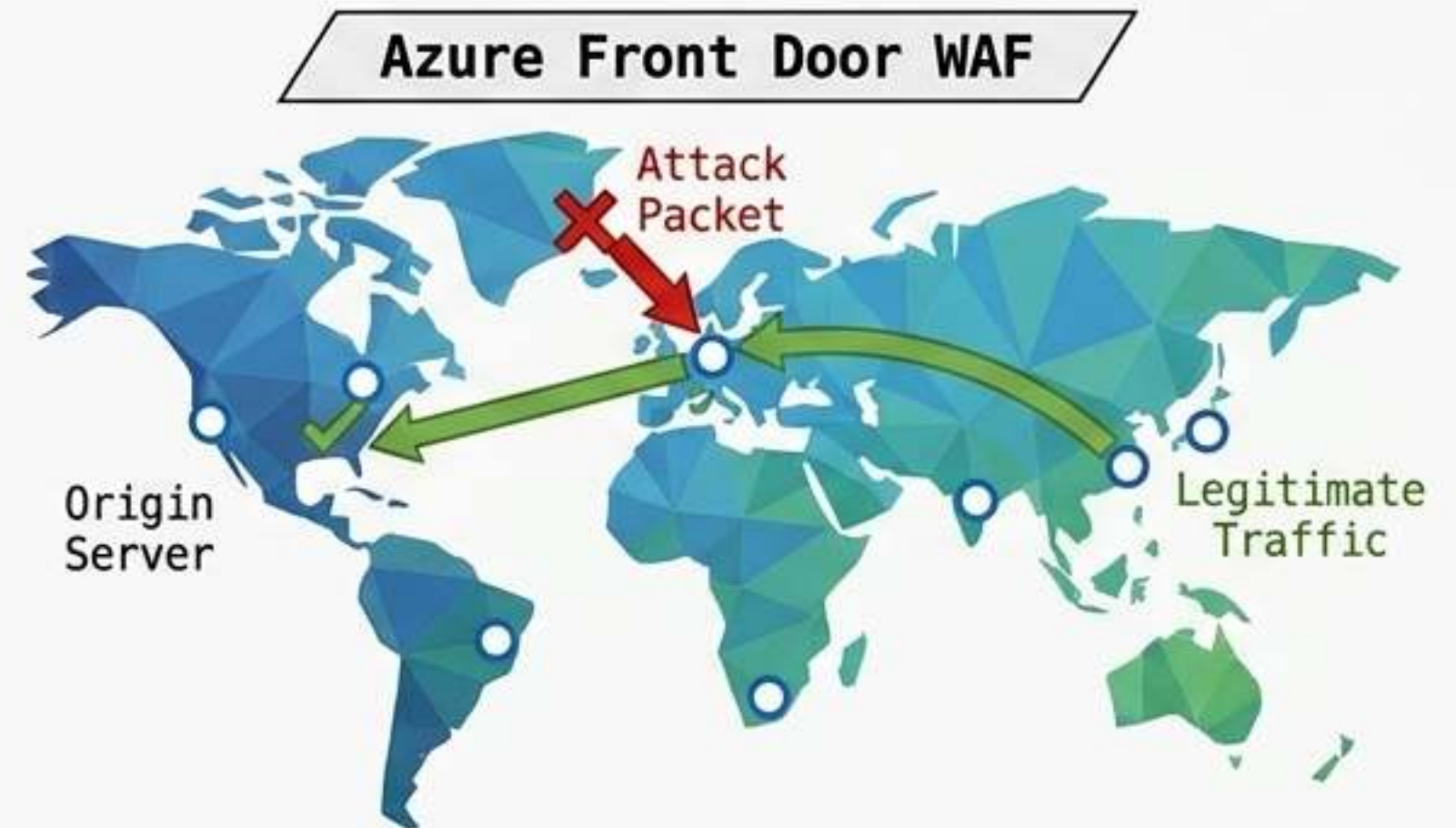
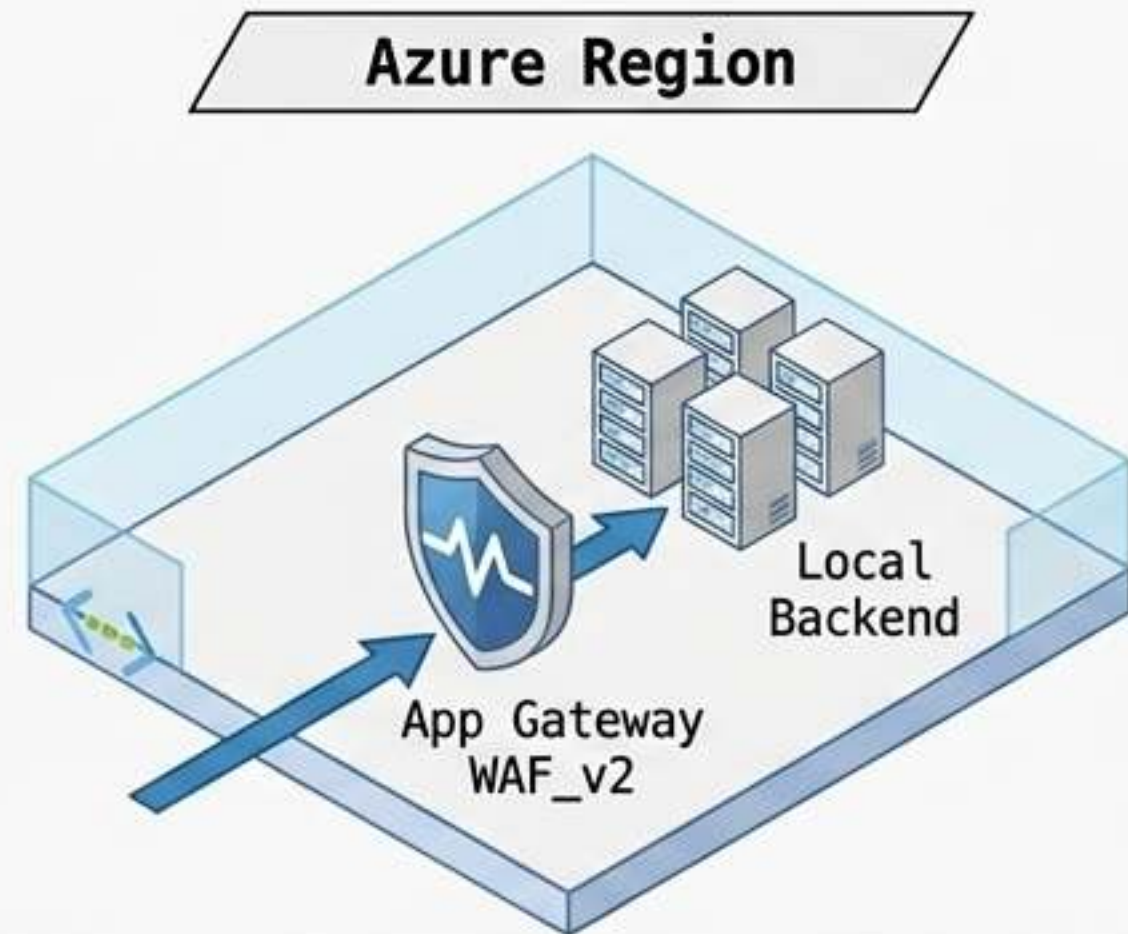
Feature	VNet Service Endpoint	Private Endpoint (Private Link)
Address Space	PaaS keeps its Public IP.	Gets a dedicated Private IP inside your VNet.
DNS Resolution	Resolves to Public IP.	Resolves to Private IP via privatelink.* Private DNS Zone.
Exfiltration Risk	Medium - Compromised VM can still reach attacker-owned PaaS in another tenant.	Low - Connection is mapped to one specific sub-resource instance.
Public Exposure	Endpoint remains on the internet.	Public access can be 100% disabled.

Layer 4 Routing vs. Layer 7 Deep Inspection



Feature	Network Security Group (NSG)	Azure Firewall
Inspection Layer	L3/L4 - Source/Destination IP, Port, Protocol.	Up to L7 - FQDN, Application rules.
Statefulness	Evaluated per rule priority, manual return rules sometimes needed.	Fully stateful, tracks return traffic automatically.
Management Boundary	Decentralized, applied at Subnet or NIC level.	Centralized via Firewall Policies and Azure Firewall Manager.
Threat Intelligence	None, static rules.	Active, automatically Alerts & Denies traffic to known malicious IPs.

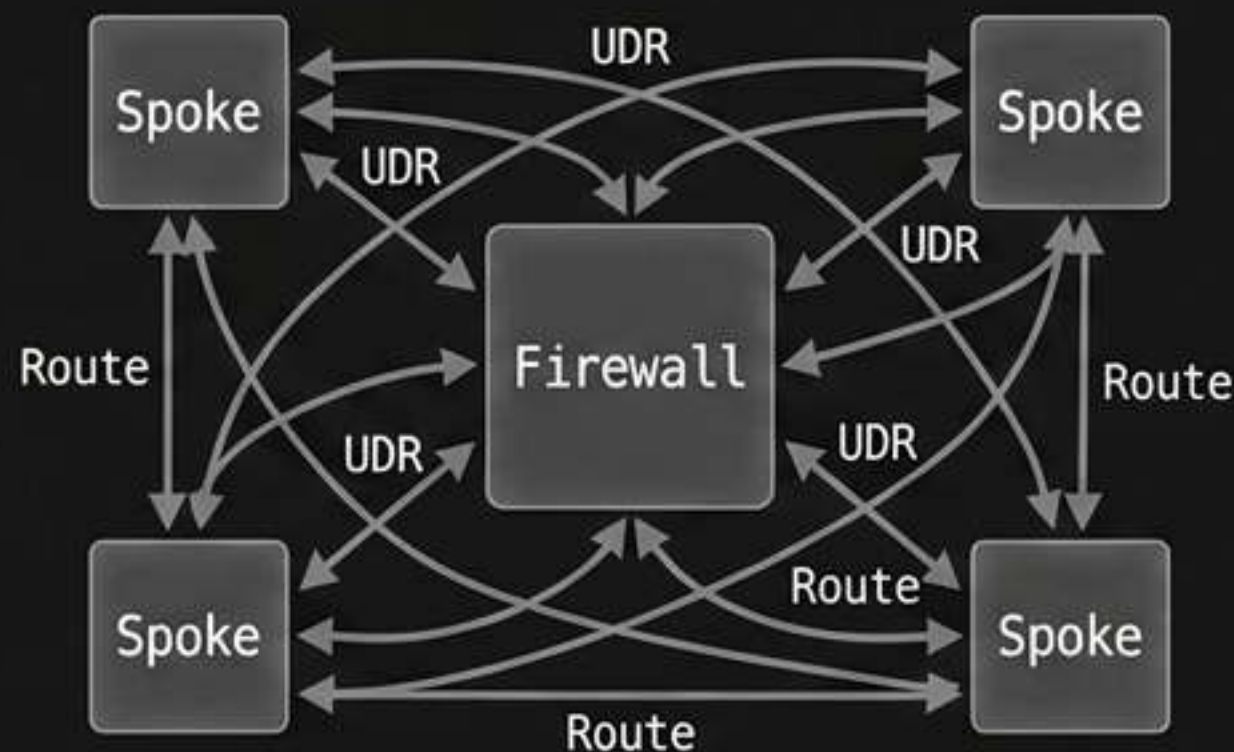
Intercepting Web Attacks at the Proper Edge



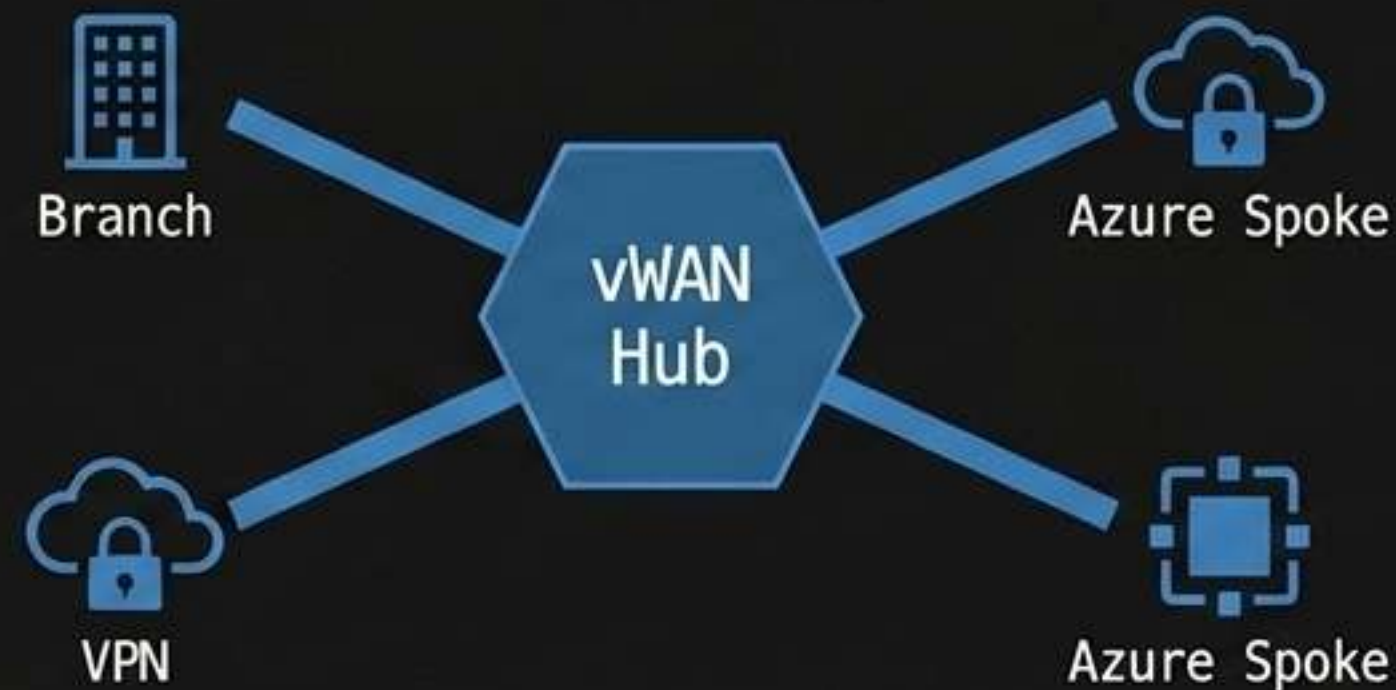
Feature	Application Gateway WAF_v2	Azure Front Door WAF
Deployment Boundary	Regional – Deployed inside your specific VNet.	Global – Deployed at Microsoft's global edge network.
Topology Fit	Best for single-region apps requiring deep VNet integration.	Best for globally distributed, multi-region apps.
Protection Focus	Deep VNet isolation.	Catching DDoS and Layer 7 attacks far from the origin infrastructure.

Manual Routing vs. Automated Hub Topologies

Hub-and-Spoke VNet
(Manual/Chaotic)



Secured Virtual Hub / vWAN
(Automated/Organized)



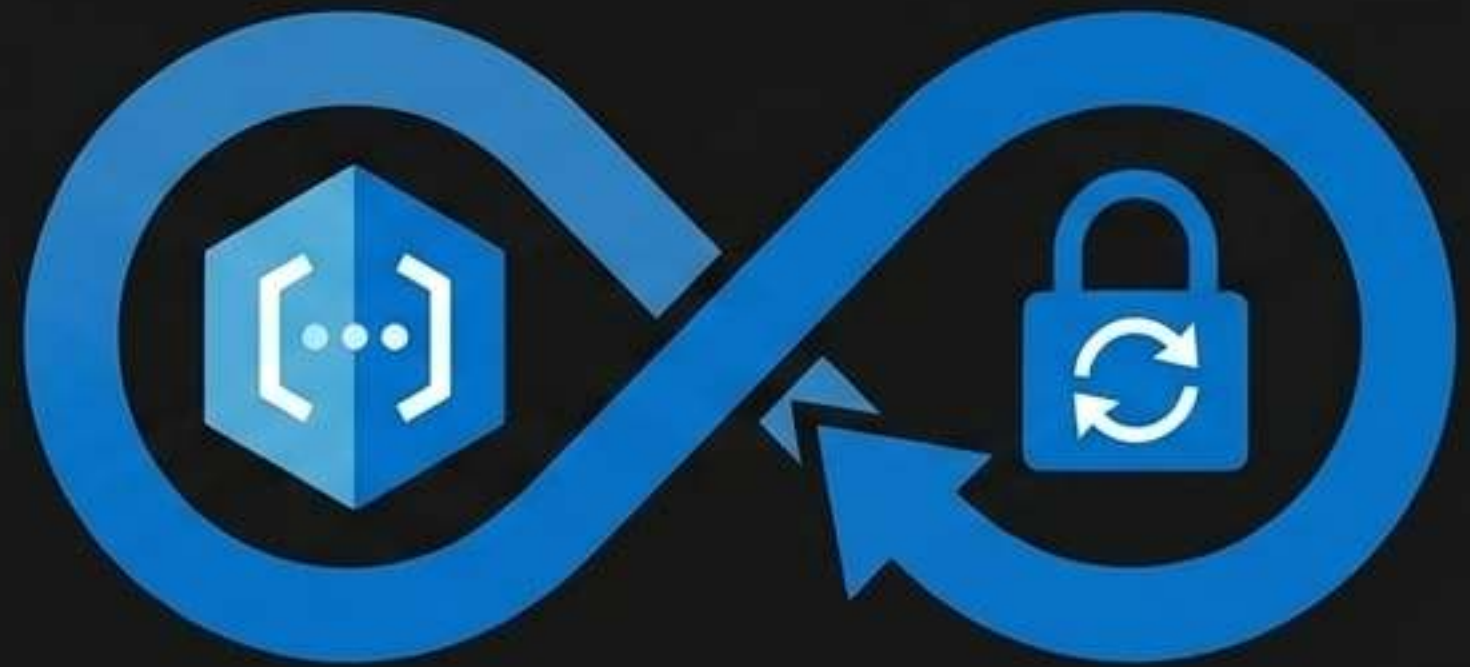
Feature	Hub-and-Spoke VNet	Secured Virtual Hub / vWAN
Route Management	Manual – UDRs required on every spoke subnet.	Automated – Routing Intent seamlessly pushes routes.
Branch & VPN Integration	Requires hand-built VPN/ExpressRoute Gateways.	Native connectivity baked into Virtual WAN.
Multi-Region Scale	Complex manual peering required per region.	Native – vWAN automatically manages hub-to-hub routing globally.

The Lifecycle of an Application Secret

Service Principal

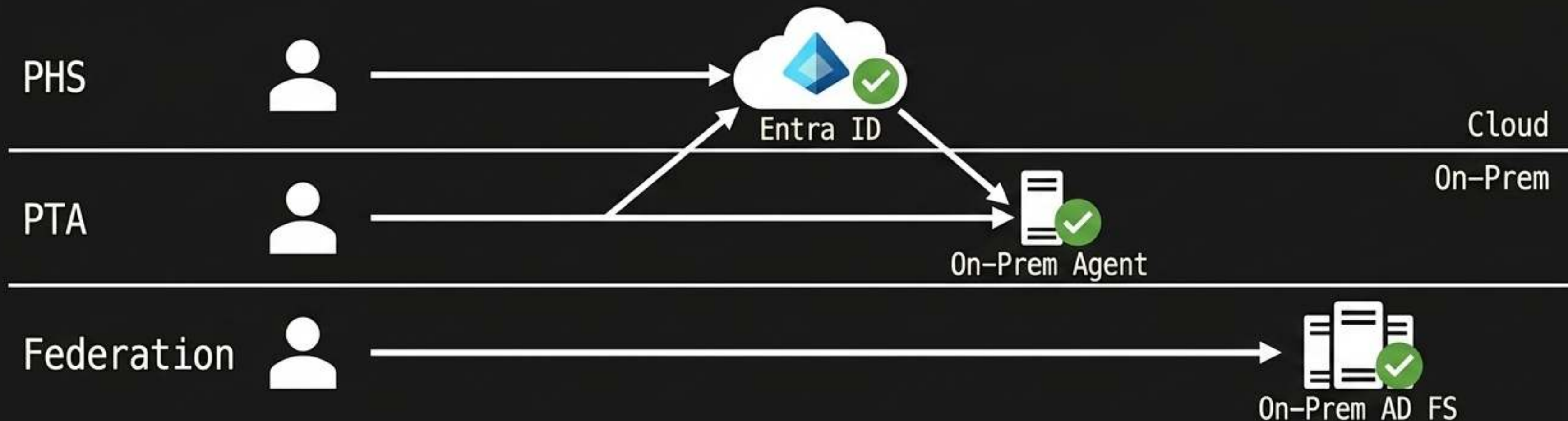


Managed Identity



Feature	Service Principal (App Registration)	Managed Identity
Credential Lifecycle	Manual – You must create, store, and rotate secrets/certificates.	Automated – Entra ID securely rotates the credentials behind the scenes.
Resource Binding	Independent – Exists indefinitely until manually deleted.	System-assigned is tied 1:1 to the Azure Resource; deleted when the resource is deleted.
Primary Use Case	Third-party apps, multi-tenant auth, complex scenarios.	Azure-to-Azure authentication, e.g., Azure Function authenticating to Key Vault.

Where the Password Check Actually Happens



Feature	Password Hash Sync (PHS)	Pass-Through Auth (PTA)	Federation (AD FS)
Auth Location	Cloud – Entra ID.	Cloud via real-time On-Prem Agent check.	Strictly On-Premises via AD FS.
On-Prem Policy Enforcement	Delayed – syncs every 2 mins.	Real-time – immediately enforces sign-in hours and	Real-time.
Infrastructure Requirement	Entra Connect Sync only – highly available.	Entra Connect + Outbound-only PTA Agents.	Heavy – AD FS Farm, WAP, Load Balancers.

Eliminating Standing Access Risk

Standing Access = Permanent Risk

Just-in-Time (JIT) Access

Feature	Active Assignment	Eligible Assignment (PIM)
Access State	"Always On" standing privileges.	Just-In-Time elevation.
Activation Process	None. Usable immediately upon login.	Requires explicit activation, justification, MFA, and/or peer approval.
Time Boundary	Indefinite or long-term.	Strict maximum duration per session, e.g., 4 hours.
Best Practice Use	Reserved exclusively for 2 "Break-Glass" emergency accounts.	Mandatory for all Global and Security Admin roles.

Governing Security Posture Across Boundaries

Azure Lighthouse

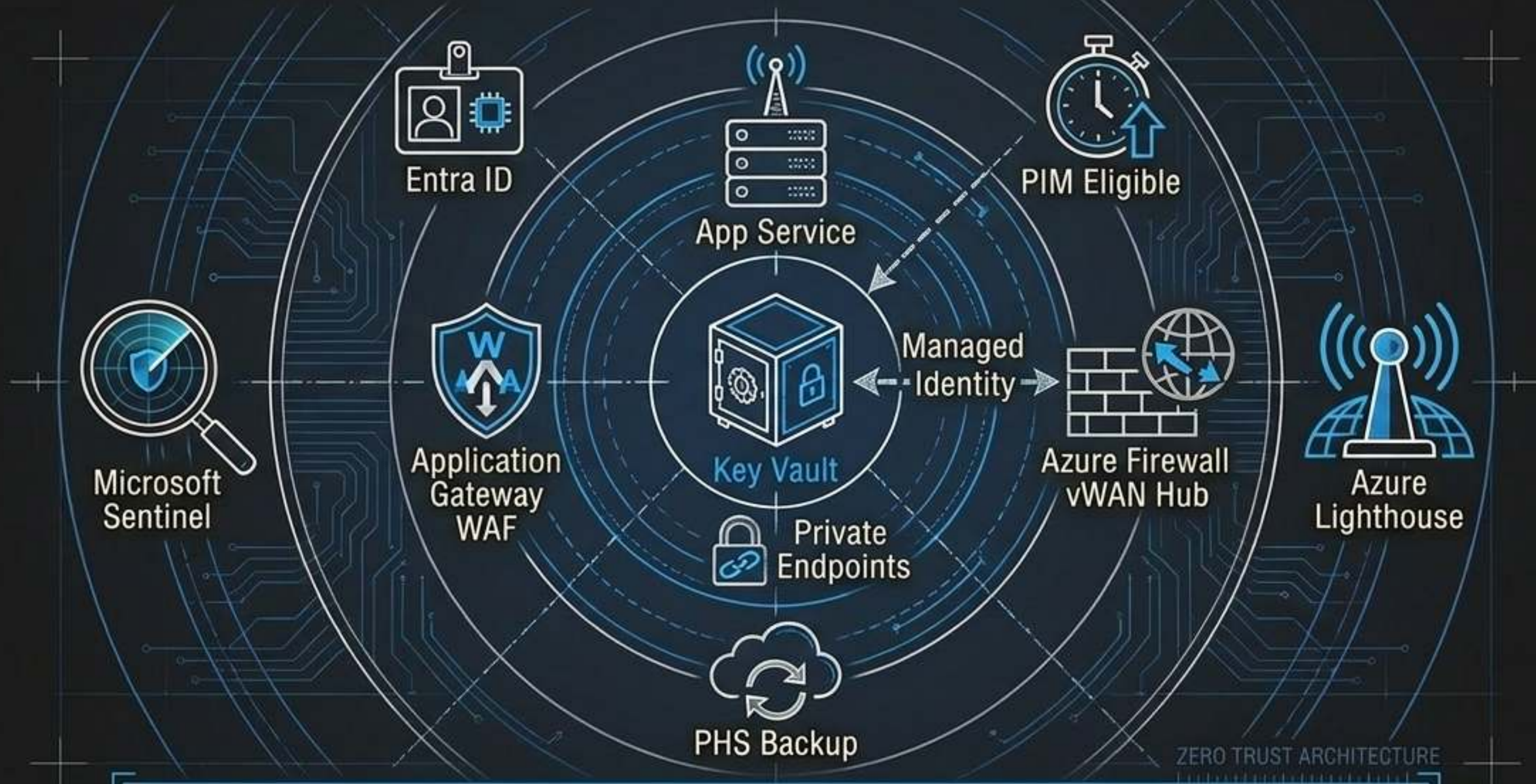


Sentinel Workspace Manager



Feature	Azure Lighthouse	Sentinel Workspace Manager
Primary Function	Cross-tenant delegated resource management.	Cross-workspace SOC content distribution.
Identity Mechanics	Uses your home tenant identity; zero B2B guest accounts required.	Operates on workspaces you already have RBAC access to.
Target Audience	Managed Security Service Providers (MSSPs).	Centralized SOC teams keeping detection rules synced globally.

The Zero Trust Production Blueprint



None of these services exist in a vacuum. A WAF is useless if a public IP bypass exists.
A Private Endpoint is useless if a compromised Global Admin has standing access.
True cloud security requires mastering the whole board.